

Practice 1

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```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
##   filter, lag  
  
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(readr)  
library(haven)
```

```
data <- haven::read_dta("nyc_schools.dta")  
# Source: New York City Department of Education records, assembled by Nathan Favero
```

Renaming Variables

```
head(data)
```

```
## # A tibble: 6 x 19  
##   dbn      schoolname      schooltype overallscore overallgrade percentilerank  
##   <chr>   <chr>           <chr>          <dbl> <chr>              <dbl>  
## 1 01M015 P.S. 015 Roberto C~ Elementary      39    C                15  
## 2 01M019 P.S. 019 Asher Levy Elementary     55.9  B                56  
## 3 01M020 P.S. 020 Anna Silv~ Elementary     40.2  C                17  
## 4 01M034 P.S. 034 Franklin ~ K-8           67.5  A                83  
## 5 01M063 P.S. 063 William M~ Elementary     59.3  B                63  
## 6 01M064 P.S. 064 Robert Si~ Elementary     48.9  C                37  
## # i 13 more variables: progressgrade <chr>, performancegrade <chr>,  
## #   environmentgrade <chr>, closingtheachievementgappoints <dbl>,  
## #   principal <chr>, enrollment <dbl>, district <dbl>, iep <chr>,  
## #   economicneedindex <dbl>, blackhispanic <chr>, ell <chr>, thgrmathela <chr>,  
## #   peerindex <dbl>
```

```
# Rename a variable
data <- data %>% rename(school = schoolname)

# See if changes made
head(data)

## # A tibble: 6 x 19
##   dbn   school schooltype overallscore overallgrade percentilerank progressgrade
##   <chr> <chr>   <chr>           <dbl> <chr>           <dbl> <chr>
## 1 01M0~ P.S. ~ Elementary      39    C              15 F
## 2 01M0~ P.S. ~ Elementary     55.9 B              56 B
## 3 01M0~ P.S. ~ Elementary     40.2 C              17 D
## 4 01M0~ P.S. ~ K-8           67.5 A              83 B
## 5 01M0~ P.S. ~ Elementary     59.3 B              63 B
## 6 01M0~ P.S. ~ Elementary     48.9 C              37 C
## # i 12 more variables: performancegrade <chr>, environmentgrade <chr>,
## #   closingtheachievementgappoints <dbl>, principal <chr>, enrollment <dbl>,
## #   district <dbl>, iep <chr>, economicneedindex <dbl>, blackhispanic <chr>,
## #   ell <chr>, thgrmathela <chr>, peerindex <dbl>
```

Summary Statistics for Subsets

Method 1

```
summary(data$overallscore[data$schooltype == "Elementary"])
```

```
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##  12.70  44.83   53.80   54.54  64.70  102.20     8
```

```
summary(data$overallscore[data$schooltype == "Middle"])
```

```
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##  13.40  45.90   55.30   55.94  65.47  102.20    25
```

```
summary(data$overallscore[data$schooltype == "K-8"])
```

```
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##  26.20  44.22   53.95   54.98  64.27   91.90     1
```

```
#Guess what will following codes generate
```

```
data$schooltype == "Elementary"
```

```
##   [1] TRUE TRUE TRUE FALSE TRUE TRUE TRUE TRUE TRUE FALSE TRUE FALSE
##  [13] FALSE FALSE FALSE TRUE FALSE FALSE TRUE TRUE TRUE FALSE FALSE FALSE
##  [25] FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
##  [37] TRUE TRUE FALSE FALSE FALSE TRUE TRUE FALSE TRUE FALSE TRUE TRUE
##  [49] FALSE TRUE TRUE TRUE FALSE FALSE TRUE FALSE FALSE FALSE FALSE TRUE
```

```

## [61] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE TRUE
## [73] FALSE TRUE TRUE TRUE FALSE TRUE FALSE TRUE FALSE FALSE TRUE TRUE
## [85] TRUE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [97] FALSE FALSE FALSE FALSE FALSE FALSE TRUE TRUE FALSE FALSE FALSE TRUE
## [109] TRUE FALSE TRUE FALSE TRUE TRUE FALSE TRUE TRUE FALSE FALSE TRUE
## [121] FALSE FALSE FALSE TRUE FALSE FALSE TRUE TRUE TRUE FALSE TRUE FALSE
## [133] TRUE FALSE TRUE TRUE FALSE TRUE TRUE FALSE TRUE TRUE FALSE FALSE
## [145] TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE TRUE TRUE FALSE
## [157] TRUE TRUE FALSE TRUE TRUE TRUE TRUE TRUE FALSE TRUE TRUE TRUE FALSE
## [169] TRUE TRUE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE
## [181] TRUE FALSE FALSE FALSE FALSE FALSE TRUE TRUE FALSE TRUE TRUE TRUE
## [193] TRUE FALSE TRUE FALSE TRUE TRUE TRUE FALSE TRUE TRUE TRUE FALSE
## [205] TRUE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE TRUE TRUE FALSE
## [217] FALSE TRUE TRUE TRUE TRUE TRUE FALSE TRUE TRUE TRUE TRUE TRUE FALSE
## [229] TRUE TRUE FALSE FALSE TRUE FALSE TRUE TRUE TRUE TRUE TRUE TRUE FALSE
## [241] FALSE FALSE TRUE TRUE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [253] FALSE FALSE FALSE FALSE TRUE FALSE TRUE TRUE TRUE TRUE TRUE TRUE
## [265] TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE TRUE FALSE TRUE FALSE
## [277] TRUE TRUE TRUE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE
## [289] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE TRUE FALSE
## [301] FALSE FALSE TRUE TRUE TRUE FALSE FALSE TRUE TRUE TRUE FALSE FALSE
## [313] TRUE TRUE TRUE TRUE TRUE FALSE TRUE TRUE TRUE TRUE TRUE FALSE
## [325] FALSE FALSE TRUE TRUE FALSE FALSE TRUE FALSE FALSE FALSE TRUE FALSE
## [337] FALSE FALSE TRUE TRUE TRUE FALSE TRUE FALSE FALSE TRUE FALSE TRUE
## [349] TRUE FALSE FALSE TRUE TRUE FALSE FALSE TRUE FALSE FALSE TRUE FALSE
## [361] TRUE TRUE TRUE TRUE TRUE FALSE TRUE FALSE TRUE TRUE TRUE TRUE
## [373] TRUE TRUE TRUE TRUE TRUE FALSE FALSE FALSE TRUE TRUE FALSE TRUE
## [385] FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [397] FALSE FALSE FALSE FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [409] TRUE FALSE TRUE FALSE TRUE TRUE FALSE TRUE TRUE FALSE FALSE FALSE
## [421] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE
## [433] FALSE FALSE FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [445] TRUE TRUE TRUE TRUE FALSE FALSE TRUE TRUE FALSE FALSE TRUE FALSE
## [457] TRUE FALSE TRUE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE
## [469] TRUE TRUE TRUE TRUE TRUE TRUE FALSE TRUE FALSE TRUE TRUE TRUE
## [481] FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE FALSE TRUE FALSE
## [493] FALSE FALSE FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE
## [505] TRUE FALSE TRUE TRUE TRUE TRUE TRUE FALSE TRUE TRUE TRUE TRUE
## [517] TRUE TRUE TRUE TRUE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [529] FALSE FALSE FALSE TRUE FALSE TRUE TRUE TRUE TRUE TRUE FALSE TRUE
## [541] FALSE TRUE TRUE FALSE FALSE FALSE TRUE TRUE FALSE FALSE FALSE FALSE
## [553] TRUE TRUE TRUE FALSE TRUE TRUE TRUE FALSE TRUE TRUE FALSE FALSE
## [565] TRUE FALSE FALSE TRUE TRUE TRUE TRUE FALSE TRUE TRUE FALSE FALSE FALSE
## [577] FALSE FALSE TRUE FALSE FALSE TRUE TRUE TRUE FALSE FALSE FALSE FALSE
## [589] FALSE FALSE FALSE FALSE TRUE TRUE TRUE TRUE FALSE TRUE TRUE FALSE
## [601] TRUE TRUE TRUE TRUE TRUE FALSE FALSE FALSE FALSE FALSE TRUE TRUE
## [613] TRUE FALSE TRUE TRUE TRUE TRUE FALSE FALSE FALSE TRUE FALSE TRUE
## [625] TRUE FALSE TRUE TRUE TRUE TRUE FALSE FALSE FALSE FALSE FALSE TRUE
## [637] TRUE FALSE FALSE FALSE TRUE FALSE FALSE TRUE FALSE TRUE TRUE FALSE
## [649] TRUE TRUE TRUE TRUE FALSE TRUE TRUE TRUE TRUE FALSE TRUE TRUE
## [661] FALSE FALSE TRUE FALSE TRUE TRUE FALSE FALSE FALSE FALSE TRUE FALSE
## [673] TRUE TRUE FALSE TRUE FALSE FALSE TRUE FALSE FALSE TRUE TRUE FALSE
## [685] TRUE TRUE TRUE TRUE TRUE FALSE TRUE TRUE TRUE FALSE FALSE FALSE
## [697] FALSE FALSE TRUE FALSE FALSE FALSE TRUE FALSE FALSE FALSE TRUE FALSE

```

```
## [709] FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE
## [721] FALSE TRUE TRUE FALSE TRUE FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [733] FALSE TRUE TRUE FALSE FALSE FALSE FALSE FALSE TRUE FALSE TRUE FALSE
## [745] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE
## [757] FALSE FALSE FALSE FALSE FALSE TRUE TRUE TRUE TRUE TRUE FALSE TRUE
## [769] FALSE TRUE TRUE FALSE FALSE TRUE FALSE TRUE TRUE TRUE FALSE FALSE
## [781] FALSE FALSE FALSE FALSE TRUE TRUE TRUE TRUE TRUE TRUE FALSE TRUE
## [793] TRUE TRUE TRUE TRUE FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [805] TRUE FALSE TRUE TRUE TRUE FALSE FALSE TRUE FALSE FALSE TRUE TRUE
## [817] TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE TRUE TRUE
## [829] TRUE TRUE FALSE FALSE TRUE TRUE TRUE TRUE FALSE TRUE TRUE FALSE
## [841] TRUE FALSE TRUE TRUE TRUE TRUE TRUE TRUE FALSE TRUE FALSE FALSE
## [853] FALSE TRUE FALSE FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [865] TRUE TRUE TRUE TRUE FALSE TRUE TRUE FALSE TRUE FALSE FALSE FALSE
## [877] TRUE FALSE TRUE FALSE FALSE FALSE TRUE TRUE FALSE FALSE TRUE TRUE
## [889] FALSE FALSE TRUE FALSE TRUE FALSE FALSE FALSE FALSE FALSE TRUE TRUE
## [901] TRUE TRUE TRUE TRUE FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [913] TRUE TRUE TRUE FALSE TRUE TRUE FALSE TRUE TRUE FALSE TRUE TRUE
## [925] FALSE TRUE FALSE FALSE FALSE FALSE TRUE TRUE TRUE TRUE TRUE TRUE
## [937] TRUE TRUE FALSE TRUE FALSE FALSE TRUE TRUE TRUE TRUE TRUE TRUE
## [949] FALSE FALSE FALSE TRUE TRUE FALSE TRUE FALSE FALSE FALSE FALSE FALSE
## [961] FALSE FALSE FALSE FALSE FALSE FALSE TRUE FALSE TRUE TRUE TRUE TRUE
## [973] TRUE TRUE FALSE TRUE TRUE FALSE TRUE FALSE FALSE FALSE FALSE FALSE
## [985] TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE TRUE FALSE FALSE TRUE
## [997] FALSE FALSE FALSE TRUE FALSE TRUE TRUE TRUE TRUE FALSE TRUE TRUE
## [1009] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE TRUE FALSE
## [1021] TRUE TRUE TRUE TRUE FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [1033] TRUE TRUE TRUE FALSE TRUE FALSE TRUE TRUE TRUE TRUE TRUE TRUE
## [1045] TRUE TRUE FALSE FALSE TRUE TRUE FALSE FALSE FALSE FALSE TRUE TRUE
## [1057] TRUE TRUE TRUE TRUE TRUE TRUE FALSE TRUE FALSE FALSE TRUE FALSE
## [1069] FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [1081] FALSE FALSE TRUE TRUE FALSE FALSE FALSE TRUE FALSE TRUE TRUE TRUE
## [1093] TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE TRUE FALSE
## [1105] FALSE FALSE TRUE TRUE FALSE FALSE TRUE TRUE FALSE FALSE FALSE FALSE
## [1117] FALSE TRUE FALSE TRUE TRUE TRUE TRUE TRUE FALSE FALSE FALSE FALSE
## [1129] TRUE FALSE TRUE TRUE FALSE FALSE FALSE TRUE TRUE TRUE TRUE FALSE
## [1141] FALSE FALSE FALSE TRUE FALSE FALSE TRUE FALSE TRUE TRUE TRUE TRUE
## [1153] TRUE TRUE TRUE TRUE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE
## [1165] TRUE TRUE
```

Breaking Down the Command:

1. **data\$overallscore:** This selects the `overallscore` column from the `data` dataframe.
2. **data\$schooltype == "Elementary":** This creates a logical vector (TRUE/FALSE) that is TRUE for rows where the `schooltype` column equals "Elementary" and FALSE otherwise.
3. **data\$overallscore[data\$schooltype == "Elementary"]:** This uses the logical vector to subset `overallscore`, selecting only those values where `schooltype` is "Elementary."
4. **summary():** The `summary()` function then computes summary statistics (such as the minimum, 1st quartile, median, mean, 3rd quartile, and maximum) for the selected subset of `overallscore`.

Method 2

```
data%>%
  filter(schooltype=="Elementary")%>%
  select(overallscore)%>%
  summary()
```

```
##  overallscore
##  Min.   : 12.70
##  1st Qu.: 44.83
##  Median : 53.80
##  Mean   : 54.54
##  3rd Qu.: 64.70
##  Max.   :102.20
##  NA's   :8
```

```
data%>%
  filter(schooltype=="Middle")%>%
  .$overallscore%>% # see the break-downs below
  summary()
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##  13.40  45.90   55.30   55.94  65.47  102.20     25
```

Breaking Down the Command:

1. **data %>%**: Starts a pipeline where the **data** dataframe is passed into the next function.
2. **filter(schooltype == "Middle")**: Filters the **data** dataframe to include only rows where the **schooltype** column is equal to "Middle." The result is a smaller dataframe containing only middle schools.
3. **.\$overallscore**: Extracts the **overallscore** column from the filtered dataframe. The **.** refers to the dataframe that results from the previous step in the pipeline.
4. **summary()**: Applies the **summary()** function to the extracted **overallscore** column, generating summary statistics such as the minimum, 1st quartile, median, mean, 3rd quartile, and maximum for middle schools.

Rescaling Variables

```
head(data)
```

```
## # A tibble: 6 x 19
##   dbn   school schooltype overallscore overallgrade percentilerank progressgrade
##   <chr> <chr>   <chr>          <dbl> <chr>                <dbl> <chr>
## 1 01M0~ P.S. ~ Elementary      39    C                    15    F
## 2 01M0~ P.S. ~ Elementary     55.9    B                    56    B
## 3 01M0~ P.S. ~ Elementary     40.2    C                    17    D
## 4 01M0~ P.S. ~ K-8           67.5    A                    83    B
```

```
## 5 01M0~ P.S. ~ Elementary      59.3 B      63 B
## 6 01M0~ P.S. ~ Elementary      48.9 C      37 C
## # i 12 more variables: performancegrade <chr>, environmentgrade <chr>,
## #   closingtheachievementgappoints <dbl>, principal <chr>, enrollment <dbl>,
## #   district <dbl>, iep <chr>, economicneedindex <dbl>, blackhispanic <chr>,
## #   ell <chr>, thgrmathela <chr>, peerindex <dbl>
```

```
# Rescale the overall score to range from 0 to 1
data <- data %>%
  mutate(overallscore = overallscore / 100)

summary(data$overallscore)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
## 0.1270  0.4497  0.5415  0.5505  0.6490  1.0220      34
```

Creating Dummy Variables for Grades

```
table(data$overallgrade)
```

```
##
##      A    B    C    D    F
## 34 287 399 348  76  22
```

```
# Create dummy variables for letter grades
data <- data %>% mutate(
  gradeA = ifelse(overallgrade == "A", 1, ifelse(overallgrade == "", NA, 0)),
  gradeB = ifelse(overallgrade == "B", 1, ifelse(overallgrade == "", NA, 0)),
  gradeC = ifelse(overallgrade == "C", 1, ifelse(overallgrade == "", NA, 0)),
  gradeD = ifelse(overallgrade == "D", 1, ifelse(overallgrade == "", NA, 0)),
  gradeF = ifelse(overallgrade == "F", 1, ifelse(overallgrade == "", NA, 0)),
  grade_NA = ifelse(overallgrade == "", 1, 0)
)

head(data)
```

```
## # A tibble: 6 x 25
##   dbn   school schooltype overallscore overallgrade percentilerank progressgrade
##   <chr> <chr>   <chr>          <dbl> <chr>              <dbl> <chr>
## 1 01M0~ P.S. ~ Elementary      0.39  C              15 F
## 2 01M0~ P.S. ~ Elementary      0.559 B              56 B
## 3 01M0~ P.S. ~ Elementary      0.402 C              17 D
## 4 01M0~ P.S. ~ K-8             0.675 A              83 B
## 5 01M0~ P.S. ~ Elementary      0.593 B              63 B
## 6 01M0~ P.S. ~ Elementary      0.489 C              37 C
## # i 18 more variables: performancegrade <chr>, environmentgrade <chr>,
## #   closingtheachievementgappoints <dbl>, principal <chr>, enrollment <dbl>,
## #   district <dbl>, iep <chr>, economicneedindex <dbl>, blackhispanic <chr>,
## #   ell <chr>, thgrmathela <chr>, peerindex <dbl>, gradeA <dbl>, gradeB <dbl>,
## #   gradeC <dbl>, gradeD <dbl>, gradeF <dbl>, grade_NA <dbl>
```

```
table(data$gradeA,data$overallgrade)
```

```
##
##           A    B    C    D    F
##    0    0    0 399 348  76  22
##    1    0 287    0    0    0    0
```

Breaking Down the Command:

- `gradeA = ifelse(overallgrade == "A", 1, ifelse(overallgrade == "", NA, 0)):`
 - `ifelse(overallgrade == "A", 1, ifelse(overallgrade == "", NA, 0)):`
 - * Checks if `overallgrade` is “A”. If true, assigns 1 to `gradeA`.
 - * If `overallgrade` is missing (empty string), assigns NA to `gradeA`.
 - * If neither condition is true, assigns 0 to `gradeA`.
 - This pattern is repeated for `gradeB`, `gradeC`, `gradeD`, and `gradeF`, where the check is for “B”, “C”, “D”, and “F”, respectively.
- `grade_missing = ifelse(overallgrade == "", 1, 0):`
 - This creates a binary indicator that assigns 1 if `overallgrade` is missing (empty string) and 0 otherwise.

Creating an Index Variable

Now, let’s create an index of the progress grade and the performance grade. We first convert the grades to numeric variables.

We assign a score of 4 to schools with an A, 3 for a B, etc.

```
# Creating an index of progress grade and performance grade
data <- data %>% mutate(
  progress = case_when(
    progressgrade == "A" ~ 4,
    progressgrade == "B" ~ 3,
    progressgrade == "C" ~ 2,
    progressgrade == "D" ~ 1,
    progressgrade == "F" ~ 0,
    TRUE ~ NA_real_
  ),
  performance = case_when(
    performancegrade == "A" ~ 4,
    performancegrade == "B" ~ 3,
    performancegrade == "C" ~ 2,
    performancegrade == "D" ~ 1,
    performancegrade == "F" ~ 0,
    TRUE ~ NA_real_
  ),
  index = (progress + performance)/2
)

head(data)
```

```
## # A tibble: 6 x 28
##   dbn   school schooltype overallscore overallgrade percentilerank progressgrade
##   <chr> <chr>   <chr>           <dbl> <chr>           <dbl> <chr>
## 1 01M0~ P.S. ~ Elementary      0.39  C              15 F
## 2 01M0~ P.S. ~ Elementary      0.559 B              56 B
## 3 01M0~ P.S. ~ Elementary      0.402 C              17 D
## 4 01M0~ P.S. ~ K-8             0.675 A              83 B
## 5 01M0~ P.S. ~ Elementary      0.593 B              63 B
## 6 01M0~ P.S. ~ Elementary      0.489 C              37 C
## # i 21 more variables: performancegrade <chr>, environmentgrade <chr>,
## #   closingtheachievementgappoints <dbl>, principal <chr>, enrollment <dbl>,
## #   district <dbl>, iep <chr>, economicneedindex <dbl>, blackhispanic <chr>,
## #   ell <chr>, thgrmathela <chr>, peerindex <dbl>, gradeA <dbl>, gradeB <dbl>,
## #   gradeC <dbl>, gradeD <dbl>, gradeF <dbl>, grade_NA <dbl>, progress <dbl>,
## #   performance <dbl>, index <dbl>
```

```
# Note: NA + 10 = ?
```

```
table(data$progressgrade,data$progress)
```

```
##
##      0    1    2    3    4
##      0    0    0    0    0
##   A    0    0    0    0 185
##   B    0    0    0 316    0
##   C    0    0 367    0    0
##   D    0 162    0    0    0
##   F 102    0    0    0    0
```

```
table(data$performancegrade,data$performance)
```

```
##
##      0    1    2    3    4
##      0    0    0    0    0
##   A    0    0    0    0 439
##   B    0    0    0 302    0
##   C    0    0 247    0    0
##   D    0  95    0    0    0
##   F  49    0    0    0    0
```

NA and NA_real_

NA

- **Type:** Generic missing value.
- **Behavior:** When used in expressions, **NA** can adapt to the expected type of the output (integer, numeric, character, etc.). For example, if you're working with a numeric vector and use **NA**, it will automatically be treated as **NA_real_**.
- **Usage:** **NA** is flexible and can be used in various contexts, including vectors of different types.

NA_real_

- **Type:** Specifically a missing value of type **double** (real numbers).
- **Behavior:** Explicitly indicates that the missing value is numeric and of type **double**. This is important when you need to ensure that the data type remains consistent, especially in functions like **mutate()** where type consistency is crucial.
- **Usage:** Typically used in numeric calculations or when creating variables where the type must be explicitly **double**.

Why Use NA_real_?

- **Type Consistency:** By explicitly using **NA_real_**, you ensure that the **elementary** variable is always treated as a numeric vector. If you used **NA** instead, R would still work correctly in this case, but using **NA_real_** makes the intent clear and avoids potential issues if the type needs to be consistent, especially in more complex operations.
 - **Prevents Implicit Type Conversion:** If the context changes or if additional types are introduced, **NA_real_** ensures that R does not implicitly convert the vector to another type, which might happen with **NA**.
-

Handling String Variables

Convert Percentage Variables to Numeric

```
# We want to know how many black or hispanic students those schools have
summary(data$blackhispanic)
```

```
##      Length      Class      Mode 
##      1166 character character
```

```
#We need to convert this variable to just numbers.
# Convert blackhispanic and ell variables to numeric

data <- data %>% mutate(
  blackhispanic = as.numeric(gsub("%", "", blackhispanic))
)

data <- data %>% mutate(
  blackhispanic = gsub("%", "", blackhispanic),
  blackhispanic = as.numeric(blackhispanic)
)
```

```
)  
  
summary(data$blackhispanic)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##      3.60   52.52   92.35   74.99   97.40  100.00
```

Breaking Down the Command:

```
mutate(blackhispanic = as.numeric(gsub("%", "", blackhispanic))):
```

- **gsub("%", "", blackhispanic)**: This function removes the percentage signs (%) from the **blackhispanic** variable. The **gsub** function replaces each occurrence of % with an empty string ("").
- **as.numeric(...)**: Converts the cleaned **blackhispanic** values (now just numbers in string form) into numeric data.
- **mutate(...)**: Creates a new version of the **blackhispanic** variable within the **data** dataframe, replacing the original values with the cleaned numeric values.

Extracting Substrings

The variable **dbn** contains the district, borough, and school number. The first 2 digits are the district number. The third digit is the borough. And the fourth through sixth digits are the school number.

```
# Extract district, borough, and school number from the dbn variable  
data <- data %>% mutate(  
  distnum = substr(dbn, 1, 2),  
  borough = substr(dbn, 3, 3),  
  schoolnum = substr(dbn, 4, 6)  
)  
  
substr("abcdef", 2, 5)
```

```
## [1] "bcde"
```

Save the data

```
# Save the cleaned data  
saveRDS(data, "nyc_schools_cleaned.RDS")
```